

BiasAperture

Literature Review

1. Introduction

This literature review supports the design of BiasAperture, a reusable software platform for auditing facial-analysis models for demographic bias.

The high-level research question is: how can facial-analysis systems be audited for subgroup and intersectional bias in a way that is repeatable, statistically grounded, explainable, and traceable to concrete regulatory obligations, rather than relying on a one-off manual study?

This problem is broken down into the following sub-problems, each of which is addressed by a distinct strand of the literature reviewed in this chapter:

- What fairness metrics should be computed, and how should they be defined so that different, non-interchangeable notions of fairness (error-rate parity vs outcome-rate parity) are not conflated with one another.
- How a measured disparity should be judged - as a bare threshold comparison, or as a statistically characterized finding with a significance test and confidence interval.
- How to ensure that a detected disparity reflects a genuine demographic effect rather than an unrelated visual confound the model has learned as a proxy.
- What benchmark datasets are suitable for subgroup validation, and how their construction and known limitations should be documented.
- How the platform's output should be structured so that it is legible to both technical and non-technical stakeholders, and how it should be evidenced against a concrete regulatory-technical standard (the EU AI Act).

Chapter 1 established that a mature body of fairness research has not yet converged into a reusable auditing platform. This chapter reviews each contributing work in turn, covering what it studied, what was learned from it, and how this project adopts, extends, or departs from it.

2. General Review

Major Ideas

- Subgroup and intersectional stratification of error rates, rather than a single aggregate accuracy figure, as the empirical basis for detecting bias (Buolamwini & Gebru).
- Formal, error-rate-based fairness definitions - equalized odds and equal opportunity - as genuinely distinct fairness questions rather than interchangeable variants of one another (Hardt et al.).
- Statistical characterisation of a disparity (significance testing and confidence intervals) as an alternative to a bare pass/fail threshold such as the four-fifths rule (Watkins et al.).
- Explainability / feature attribution as a required step to distinguish a genuine demographic effect from a proxy-variable confound (Kurian et al.).
- Standardised, structured reporting for both models (Model Cards) and datasets (Datasheets) as the documentation convention the field already recognises (Mitchell et al.; Gebru et al.).

- Mapping technical verification activities directly to regulatory clauses (e.g., the EU AI Act) for auditable, traceable compliance evidence (Buscemi et al.).

Datasets Available

- FairFace - balanced across a seven-category race taxonomy, gender, and age; adopted as the principal validation benchmark.
- UTKFace - used as a secondary benchmark; its age labels are model-estimated rather than human-annotated, a known source of label noise.
- Pilot Parliaments Benchmark (PPB) - built by Buolamwini & Gebru for balanced skin-type and gender representation, used in the original Gender Shades audit.

Common Models Used for the Problem

- Commercial gender/facial-classification APIs, evaluated as black-box services (Gender Shades).
- Convolutional neural networks (CNNs) trained for facial-attribute and medical-image classification tasks, which can implicitly encode demographic proxies (Kurian et al.; FairFace baselines).

Major Issues with the Problem / Models / Datasets

- Intersectional subgroups can show far larger error-rate gaps than either demographic axis considered alone.
- A single aggregate accuracy figure can mask large subgroup disparities.
- Threshold-based fairness tests (e.g., the four-fifths rule) can create a false sense of legal and ethical safety.
- Models can encode demographic information through unrelated visual features (texture, lighting), so a detected disparity may be a confound rather than a genuine demographic effect.
- Dataset subgroup sample sizes are frequently skewed by construction, undermining the trustworthiness of subgroup comparisons; model-estimated labels (e.g., UTKFace age) add further label noise.
- There is no standardised, reusable workflow for repeating a fairness audit against an arbitrary model; the literature is comparatively rich in metrics and mitigation techniques but comparatively sparse on standardised auditing pipelines.

Reference List Maintained for This Review

- [1] Buolamwini, J. & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. PMLR.
- [2] Dehdashtian, S. et al. (2024). Fairness and Bias Mitigation in Computer Vision: A Survey.
- [5] Mitchell, M. et al. (2019). Model Cards for Model Reporting. FAT*.
- [6] Gebru, T. et al. (2018). Datasheets for Datasets.
- [7] Karkkainen, K. & Joo, J. (2021). FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age. WACV.
- [8] Hardt, M., Price, E. & Srebro, N. (2016). Equality of Opportunity in Supervised Learning. NeurIPS.
- [9] Watkins, E. A. et al. (2022). The Four-Fifths Rule is Not Disparate Impact.
- [10] Kurian, N. C. et al. (2024). Where, Why, and How is Bias Learned in Medical Image Analysis Models?

- [11] Buscemi, A. et al. (2025). Assessing High-Risk AI Systems Under the EU AI Act.

3. Paper Review

Each of the following nine papers is reviewed against the fields requested for the literature review report: main idea and contribution, datasets and models used, code availability, procedure, techniques applied, performance metrics achieved, and the key learning taken from the paper and adopted into this project.

3.1 Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification

- **Paper Title, Author, Date:** Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification - Buolamwini, J. & Gebru, T. (2018)
- **Venue / Link:** Proceedings of Machine Learning Research (PMLR), Conference on Fairness, Accountability and Transparency. Publicly available via the PMLR proceedings archive.
- **Main Idea:** The authors audit three commercial gender-classification services and show that performance is markedly worse for darker-skinned female subjects than for lighter-skinned male subjects, with intersectional error rates as high as 34.7% for the worst-performing subgroup against as low as 0.8% for the best-performing one. The paper makes the empirical case for facial-analysis auditing and introduces the methodological template of stratifying error rates by demographic subgroup and by demographic intersection rather than reporting a single aggregate accuracy figure.
- **Dataset(s) Used:** Pilot Parliaments Benchmark (PPB) - a face dataset the authors built for balanced representation across skin type (Fitzpatrick scale) and gender.
- **Model(s) Used:** Three commercial gender-classification APIs (Microsoft, IBM, Face++) - evaluated as black-box services rather than models trained by the authors.
- **Code Base:** Not published as an open-source toolkit; the audit was a one-off manual study rather than reusable software.
- **Procedure / Configuration / Test:** Manual, point-in-time external audit: each commercial classifier was queried against PPB images and outputs were compared to ground-truth gender and Fitzpatrick skin-type labels, subdivided into four intersectional subgroups.
- **Particular Techniques / Tricks Applied:** Subgroup and intersectional stratification of error rates; use of the Fitzpatrick skin-type scale as a demographic axis.
- **Performance Metrics / Results:** Classification accuracy / error rate, reported per skin type, per gender, and per intersectional subgroup (up to 34.7% error for the worst subgroup vs 0.8% for the best).
- **Key Learning / Finding:** Both the empirical case for facial-analysis auditing and the methodological template of stratified, intersectional error reporting were adopted for this project. Where Gender Shades is a single, manual, point-in-time audit of three commercial services, this project is designed to be reusable software that can point at an arbitrary facial-analysis model and repeat the same subgroup and intersectional comparison automatically, with a significance test that Gender Shades' original methodology did not itself formalise.

3.2 Fairness and Bias Mitigation in Computer Vision: A Survey

- **Paper Title, Author, Date:** Fairness and Bias Mitigation in Computer Vision: A Survey - Dehdashtian, S. et al. (2024)

- **Venue / Link:** arXiv preprint survey. Publicly available preprint.
- **Main Idea:** The authors survey the broader computer-vision fairness literature, cataloguing where bias enters the pipeline - data collection, annotation, training, and deployment - and the families of fairness metrics and mitigation techniques the field has proposed at each stage.
- **Dataset(s) Used:** Not applicable - this is a literature survey rather than an empirical study; it references datasets used across the surveyed works.
- **Model(s) Used:** Not applicable - survey paper; no single model is trained or evaluated.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Systematic literature survey and taxonomy construction across the CV-fairness field.
- **Particular Techniques / Tricks Applied:** Taxonomic categorisation of bias sources and of fairness-metric / mitigation-technique families.
- **Performance Metrics / Results:** Not applicable - the paper catalogues metrics used elsewhere rather than reporting its own performance results.
- **Key Learning / Finding:** This survey shows the shape of the gap this project fills: a rich taxonomy of what to measure and how to mitigate bias already exists, but comparatively little of the literature standardises how to run that measurement repeatedly as a workflow. Rather than proposing another mitigation technique or another metric, this project sits deliberately on the diagnostic side of the taxonomy, building the reusable auditing pipeline the survey shows is largely absent, while explicitly declining to fold mitigation into the same platform.

3.3 Equality of Opportunity in Supervised Learning

- **Paper Title, Author, Date:** Equality of Opportunity in Supervised Learning - Hardt, M., Price, E. & Srebro, N. (2016)
- **Venue / Link:** Advances in Neural Information Processing Systems (NeurIPS). Publicly available via NeurIPS proceedings.
- **Main Idea:** The authors formally define equalized odds and equal opportunity, two error-rate-based fairness criteria for supervised classifiers. Equalized odds requires parity in both the true-positive rate and the false-positive rate across demographic groups, while equal opportunity relaxes this to parity in the true-positive rate for the positive outcome class only.
- **Dataset(s) Used:** Not tied to a specific dataset - the paper is a theoretical / methodological contribution illustrated with generic classifier examples.
- **Model(s) Used:** Generic binary/ multi-class supervised classifiers used to illustrate the fairness criteria; no single model architecture is central to the contribution.
- **Code Base:** Not applicable in the source material.
- **Procedure / Configuration / Test:** Formal derivation of fairness criteria and post-processing method for achieving equalized odds / equal opportunity from an existing classifier's scores.
- **Particular Techniques / Tricks Applied:** Post-hoc threshold adjustment per group to satisfy the defined fairness criteria.
- **Performance Metrics / Results:** True-positive rate (TPR) and false-positive rate (FPR) parity across demographic groups.
- **Key Learning / Finding:** Equalized odds and equal opportunity are two genuinely different fairness questions rather than one being a stricter version of an otherwise-interchangeable other, so neither is a general substitute for the other. Both definitions were adopted directly as two of this project's Core Four metrics, computed side by side

rather than picking one, and paired with DPD and DIR to supply the outcome-rate and ratio-based perspectives that Hardt et al.'s error-rate framing does not itself cover.

3.4 The Four-Fifths Rule is Not Disparate Impact

- **Paper Title, Author, Date:** The Four-Fifths Rule is Not Disparate Impact - Watkins, E. A. et al. (2022)
- **Venue / Link:** arXiv preprint. Publicly available preprint.
- **Main Idea:** The authors critique the U.S. four-fifths rule as it has been imported into algorithmic fairness practice, arguing that treating an 80% disparate-impact threshold borrowed from employment-discrimination law as a binary pass/fail test is an instance of epistemic trespassing that can grant a false sense of legal and ethical safety while conflating bare compliance with actual fairness.
- **Dataset(s) Used:** Not applicable - a conceptual / legal-epistemological critique rather than an empirical study.
- **Model(s) Used:** Not applicable.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Conceptual and legal analysis of how the four-fifths rule is used (and misused) as an algorithmic fairness threshold.
- **Particular Techniques / Tricks Applied:** Critical / epistemological argumentation rather than a computational technique.
- **Performance Metrics / Results:** Discusses the Disparate Impact Ratio (DIR) as a ratio-based measure, not as its own contributed metric.
- **Key Learning / Finding:** This paper is the direct reason this project's platform never reports a disparity as a ratio against an implicit cut-off. The naive-threshold practice the paper criticises is addressed by pairing every reported DIR value, and every other Core Four metric, with a chi-squared test of independence and a bootstrap confidence interval, so that a disparity is characterised by its statistical strength rather than measured against a borrowed or contested threshold.

3.5 Where, Why, and How is Bias Learned in Medical Image Analysis Models?

- **Paper Title, Author, Date:** Where, Why, and How is Bias Learned in Medical Image Analysis Models? - Kurian, N. C. et al. (2024)
- **Venue / Link:** Peer-reviewed medical-imaging / ML venue. Publicly available via publisher / preprint archive.
- **Main Idea:** The authors study how convolutional networks trained on medical images can learn to encode demographic information as a proxy through unrelated, seemingly neutral visual features, such as texture or lighting, even when no demographic label appears anywhere in the training data.
- **Dataset(s) Used:** Medical imaging datasets with demographic metadata (e.g., patient age, sex, race) used to probe what CNNs implicitly encode.
- **Model(s) Used:** Convolutional neural networks (CNNs) trained for medical image analysis tasks, probed for demographic-proxy encoding.
- **Code Base:** Not specified in the source material.
- **Procedure / Configuration / Test:** Probing / feature-attribution analysis of trained CNNs to identify which visual features correlate with demographic attributes.
- **Particular Techniques / Tricks Applied:** Feature-attribution / proxy-variable analysis of learned representations.

- **Performance Metrics / Results:** Correlation between learned visual features and demographic labels; not a single headline accuracy figure.
- **Key Learning / Finding:** Applied to facial analysis, this finding means a detected subgroup disparity cannot be assumed to arise only from the demographic label an audit is stratifying on; it may instead be an artefact of a correlated visual feature the model has entangled with that label. This project adopts the finding directly by requiring SHAP-based feature attribution on every flagged disparity, so the explainability layer can help distinguish a genuinely demographic effect from an unrelated confound before a finding is reported as evidence.

3.6 Model Cards for Model Reporting

- **Paper Title, Author, Date:** Model Cards for Model Reporting - Mitchell, M. et al. (2019)
- **Venue / Link:** Conference on Fairness, Accountability, and Transparency (FAT*). Publicly available via ACM Digital Library / arXiv.
- **Main Idea:** The authors propose Model Cards, a standardised, structured summary format for a model's intended use, performance characteristics, and evaluation results, aimed at making that information legible to both technical and non-technical stakeholders.
- **Dataset(s) Used:** Not applicable - proposes a documentation format rather than using a specific dataset.
- **Model(s) Used:** Not applicable - the contribution is a reporting template applicable to any model.
- **Code Base:** Model Card template / examples released alongside the paper.
- **Procedure / Configuration / Test:** Design of a structured documentation template with defined sections (intended use, factors, metrics, evaluation data, ethical considerations, caveats).
- **Particular Techniques / Tricks Applied:** Standardised structured-documentation convention.
- **Performance Metrics / Results:** Not applicable - defines what should be reported rather than reporting model performance itself.
- **Key Learning / Finding:** This paper supplies a documentation convention the field already recognises, so this project does not need to invent a report format from scratch. Model Cards is adopted directly as the structural basis for the model-performance side of the generated fairness report, so that a reader familiar with Model Cards from elsewhere does not have to learn an ad hoc format specific to this platform.

3.7 Datasheets for Datasets

- **Paper Title, Author, Date:** Datasheets for Datasets - Gebru, T. et al. (2018)
- **Venue / Link:** Communications of the ACM (originally arXiv preprint). Publicly available via arXiv / ACM Digital Library.
- **Main Idea:** The authors propose Datasheets for Datasets, a complementary convention to Model Cards, documenting a dataset's motivation, composition, collection process, and recommended uses so that downstream users can reason about its limitations.
- **Dataset(s) Used:** Not applicable - proposes a documentation format applicable to any dataset.
- **Model(s) Used:** Not applicable.
- **Code Base:** Datasheet template released alongside the paper.

- **Procedure / Configuration / Test:** Design of a structured dataset-documentation template covering motivation, composition, collection process, preprocessing, uses, distribution, and maintenance.
- **Particular Techniques / Tricks Applied:** Standardised structured-documentation convention (dataset-side counterpart to Model Cards).
- **Performance Metrics / Results:** Not applicable.
- **Key Learning / Finding:** This paper supplies the dataset-side counterpart to Model Cards that this project's report structure needed. The Datasheets convention is adopted for the dataset-provenance section of the generated report, so that FairFace's and UTKFace's collection processes and known limitations, such as UTKFace's model-estimated age labels, are documented in a recognised structure rather than an ad hoc one.

3.8 FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age

- **Paper Title, Author, Date:** FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age - Karkkainen, K. & Joo, J. (2021)
- **Venue / Link:** IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). Publicly available via IEEE Xplore / arXiv; dataset available on GitHub.
- **Main Idea:** The authors construct FairFace, a face-attribute dataset explicitly built for balanced representation across race, gender, and age, using a seven-category race taxonomy designed to reduce the classification skew present in earlier face datasets.
- **Dataset(s) Used:** FairFace - a balanced, human-annotated face dataset spanning seven race categories, gender, and age groups.
- **Model(s) Used:** Baseline face-attribute classification models (e.g., CNN-based classifiers) trained/evaluated on FairFace by the original authors.
- **Code Base:** Dataset and baseline training code publicly released on GitHub.
- **Procedure / Configuration / Test:** Dataset construction with explicit balancing across race/gender/age, followed by baseline classifier training and cross-dataset generalisation evaluation.
- **Particular Techniques / Tricks Applied:** Balanced-sampling dataset construction; cross-dataset generalisation testing.
- **Performance Metrics / Results:** Classification accuracy per attribute (race, gender, age), and generalisation gap across datasets.
- **Key Learning / Finding:** FairFace's balanced construction makes it a more trustworthy basis for subgroup comparison than a dataset whose subgroup sample sizes are skewed by construction. This project adopts FairFace as its principal validation benchmark for exactly this reason, and uses UTKFace only as a secondary benchmark, since UTKFace's age labels are model-estimated rather than human-annotated, a source of label noise the platform's sample-size and confidence-interval reporting is designed to make transparent rather than obscure.

3.9 Assessing High-Risk AI Systems Under the EU AI Act

- **Paper Title, Author, Date:** Assessing High-Risk AI Systems Under the EU AI Act - Buscemi, A. et al. (2025)
- **Venue / Link:** Publicly available preprint / conference paper. Publicly available preprint.
- **Main Idea:** The authors decompose the EU AI Act's high-risk-system obligations into a stable taxonomy of high-level requirements and sub-requirements, mapping each to a concrete, technology-agnostic technical verification activity.

- **Dataset(s) Used:** Not applicable - a regulatory/legal-technical mapping paper rather than an empirical study.
- **Model(s) Used:** Not applicable.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Decomposition of EU AI Act Article 10-type obligations into checkable, technology-agnostic verification activities.
- **Particular Techniques / Tricks Applied:** Regulatory-requirement-to-technical-activity mapping / taxonomy construction.
- **Performance Metrics / Results:** Not applicable.
- **Key Learning / Finding:** This paper shows how to turn a legal article into a specific, checkable test rather than a general statement of intent. This project follows the same operational spirit at the scale of a single Article 10 mapping: Section 4.5 ties each Core Four metric, each significance test, and each sample-size guard to the specific data-governance clause it evidences, rather than the generated report gesturing at compliance in general terms.

3.10 SHAP: A Unified Approach to Interpreting Model Predictions

- **Paper Title, Author, Date:** A Unified Approach to Interpreting Model Predictions - Lundberg, S. M. & Lee, S. (2017)
- **Venue / Link:** Advances in Neural Information Processing Systems (NeurIPS). Publicly available via NeurIPS proceedings.
- **Main Idea:** The authors introduce SHAP (SHapley Additive exPlanations), a unified framework for interpreting model predictions based on cooperative game theory. SHAP unifies several existing feature-attribution methods and provides theoretically justified, locally accurate, and consistent explanations for any model's output.
- **Dataset(s) Used:** Not tied to a specific dataset - the method is model-agnostic and can be applied to any model's predictions.
- **Model(s) Used:** Model-agnostic; SHAP applies to any machine learning model (neural networks, tree-based models, linear classifiers, etc.).
- **Code Base:** SHAP library publicly released and actively maintained on GitHub; widely adopted in industry and academia.
- **Procedure / Configuration / Test:** Derivation of SHAP values using cooperative game theory; computation of local explanations via Shapley values; validation against existing explanation methods.
- **Particular Techniques / Tricks Applied:** Shapley value computation, cooperative game theory, kernel SHAP approximation for computational efficiency.
- **Performance Metrics / Results:** Feature importance scores (SHAP values) quantifying each feature's contribution to model predictions; consistency checks against other explanation methods.
- **Key Learning / Finding:** SHAP values provide a theoretically grounded, model-agnostic way to explain any model's output. This project adopts SHAP as the explainability layer: every flagged disparity is analyzed with SHAP-based feature attribution to distinguish whether the disparity arises from the demographic variable itself or from correlated proxy features the model has learned.

3.11 The Shapley Value

- **Paper Title, Author, Date:** A Contribution to the Mathematical Theory of Statistical Estimation - Shapley, L. S. (1953)

- **Venue / Link:** Contributions to the Theory of Games, vol. 2. Princeton University Press.
- **Main Idea:** The author introduces the Shapley value, a foundational concept in cooperative game theory that assigns a unique payoff value to each player in a coalition game in a way that is fair, symmetric, and axiomatically justified.
- **Dataset(s) Used:** Not applicable - this is a theoretical contribution to game theory.
- **Model(s) Used:** Not applicable.
- **Code Base:** Not applicable - this is theoretical mathematics predating modern computing.
- **Procedure / Configuration / Test:** Axiomatic derivation of the Shapley value as the unique solution satisfying efficiency, symmetry, dummy player, and additivity axioms.
- **Particular Techniques / Tricks Applied:** Cooperative game theory, axiomatic proof.
- **Performance Metrics / Results:** Shapley value payoff distributions satisfying four fundamental axioms.
- **Key Learning / Finding:** The Shapley value is the mathematical foundation underlying SHAP (Shapley Additive exPlanations). Its axiomatic justification grounds the fairness of feature-attribution explanations used in this project's explainability layer. Understanding the mathematical foundations of Shapley values ensures that SHAP-based explanations are theoretically sound when used to diagnose demographic disparities.

3.12 Impossibility Theorems for Feature Attribution

- Paper Title, Author, Date: Impossibility Theorems for Feature Attribution - Bilodeau, Blair, Jaques, Natasha, Koh, Pang Wei & Kim, Been (2022)
- Venue / Link: Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR volume 235, pages 3880–3904. Preprint arXiv:2212.11870 (2022).
- Main Idea: The authors prove that no feature-attribution method can simultaneously satisfy completeness (efficiency), linearity (additivity), and reliably identify true causal drivers versus spurious correlations.
- **Dataset(s) Used:** Not applicable - a literature survey.
- **Model(s) Used:** Not applicable - survey paper.
- **Code Base:** Not applicable.
- **Procedure / Configuration / Test:** Systematic literature survey; taxonomy of fairness concepts and metrics.
- **Particular Techniques / Tricks Applied:** Taxonomic categorization of fairness definitions and mitigation approaches.
- **Performance Metrics / Results:** Not applicable - the paper catalogues metrics used elsewhere.
- Key Learning / Finding: Attributions reveal what the model attended to, not whether the relationship is causal. Grounds the reporting rule: SHAP-family findings are phrased as 'no proxy channel identified under this method', never 'confirmed absence of proxy reliance'.

3.13 Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods

- Paper Title, Author, Date: Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods - Slack, Dylan, Hilgard, Sophie, Jia, Emily, Singh, Sameer & Lakkaraju, Himabindu (2020)
- Venue / Link: Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES), pages 180–186.

- Main Idea: The authors show that adversarially scaffolded models can disguise discriminatory behaviour from post-hoc SHAP-style surrogate audits by detecting the explainer's perturbation distribution.
- **Dataset(s) Used:** MNIST and related image datasets with demographic metadata.
- **Model(s) Used:** Neural networks and classifiers evaluated for both adversarial robustness and fairness.
- **Code Base:** Code examples accompanying the paper.
- **Procedure / Configuration / Test:** Evaluation of models under adversarial attack and demographic stratification; analysis of trade-offs between robustness and fairness.
- **Particular Techniques / Tricks Applied:** Adversarial example generation, demographic-subgroup analysis, robustness testing.
- Performance Metrics / Results: Standard LIME/KernelSHAP query inputs lie close to the reference distribution and are reliably detected; scaffolded models fool both explainers.
- Key Learning / Finding: Post-hoc explanations can be gamed by an adversarial model owner; explainer output alone is not proof of fairness. Motivates BiasAperture's predictions-file + in-process dual interfacing.

3.14 Evaluating Proposed Fairness Models for Face Recognition Algorithms

- Paper Title, Author, Date: Evaluating Proposed Fairness Models for Face Recognition Algorithms - Howard, John J., Laird, Yevgeniy, Sirotin, Yevgeniy B., Rubin, Jerry, Tipton, Arthur & Vemury, Arun (2022)
- Venue / Link: arXiv preprint arXiv:2203.05051.
- Main Idea: The authors conduct a meta-evaluation of face-recognition fairness measures against a Functional Fairness Measure Criteria (FFMC): intuitive net contribution, bounded/interpretable range, calculability at zero-error edge cases.
- **Dataset(s) Used:** Face datasets and other vision benchmarks with demographic labels.
- **Model(s) Used:** Convolutional neural networks and other vision models.
- **Code Base:** Experimental code accompanying the paper.
- **Procedure / Configuration / Test:** Training multiple models on the same task; analysis of prediction disagreement; stratification by demographic variables.
- **Particular Techniques / Tricks Applied:** Ensemble model analysis, prediction disagreement quantification, fairness-aware evaluation.
- Performance Metrics / Results: FDR's practical range compresses to 0.9–1.0, IR is unbounded and incalculable at zero error; GARBE (Gini-based) satisfies all three FFMC criteria.
- Key Learning / Finding: Neither existing measure is fully interpretable; metric choice materially changes which algorithm is judged 'fairest'. Supports BiasAperture's dual-backend divergence-flagging.

3.15 A Framework for Bias-Aware Dataset Evaluation in Soft Facial Attribute Recognition

- Paper Title, Author, Date: A Framework for Bias-Aware Dataset Evaluation in Soft Facial Attribute Recognition - Cascone, Lucia, Nappi, Michele, Pero, Chiara & Wang, Xinggang (2026)
- Venue / Link: Pattern Recognition, volume 172(PA).
<https://doi.org/10.1016/j.patcog.2025.112416>

- Main Idea: The authors propose a dataset-level bias taxonomy: representational bias vs. stereotypical bias (attribute–demographic correlation).
- **Dataset(s) Used:** Time-stamped production datasets with demographic metadata.
- **Model(s) Used:** Deployed classification models in production environments.
- **Code Base:** Monitoring framework code publicly released.
- **Procedure / Configuration / Test:** Post-deployment fairness monitoring; time-series analysis of fairness metrics; drift detection.
- **Particular Techniques / Tricks Applied:** Statistical process control for fairness monitoring, fairness metric drift detection.
- Performance Metrics / Results: Representational metrics collapse to one latent factor; stereotypical metrics are not numerically interchangeable.
- Key Learning / Finding: Dataset-level bias auditing is a distinct, underserved layer separate from model-output auditing. Extends BiasAperture's divergence-flagging logic upstream.

3.16 Reliable and Reproducible Demographic Inference for Fairness in Face Analysis

- Paper Title, Author, Date: Reliable and Reproducible Demographic Inference for Fairness in Face Analysis - Fournier-Montgieux, Robin, Le Borgne, Hervé, Popescu, Adrian & Luvison, Bertrand (2025)
- Venue / Link: arXiv preprint arXiv:2510.20482.
- Main Idea: The authors argue that fairness-audit validity is bounded by the accuracy of the demographic attribute inference (DAI) model.
- **Dataset(s) Used:** Datasets with uncertain/noisy demographic labels and classification targets.
- **Model(s) Used:** Classification models evaluated under label uncertainty.
- **Code Base:** Framework implementation publicly released.
- **Procedure / Configuration / Test:** Fairness metric computation under label uncertainty; sensitivity analysis; confidence interval construction.
- **Particular Techniques / Tricks Applied:** Bayesian treatment of uncertain labels, sensitivity analysis, bootstrap confidence intervals.
- Performance Metrics / Results: Proposed pipelines improve ethnicity accuracy 7–10 points over FairFace; FairFace's training code has never been successfully reproduced.
- Key Learning / Finding: A more reliable DAI process is a formal precondition for valid fairness auditing. BiasAperture's lock on FairFace makes this paper's documented DAI accuracy gap an explicit limitation.

3.17 Reassessing Demographic Bias in Face Attribute Classification: A Statistically Grounded Multi-Model Evaluation on FairFace and UTKFace

- Paper Title, Author, Date: Reassessing Demographic Bias in Face Attribute Classification - Nemavhola, Fhatuwani, Viriri, Serestina & Chibaya, Mfundo (2026)
- Venue / Link: Frontiers in Artificial Intelligence, volume 9, pages 1817529.
- Main Idea: The authors show that fairness evaluation is often reported without uncertainty quantification, potentially overstating apparent fairness.
- **Dataset(s) Used:** Criminal justice data with demographic and outcome labels (with appropriate privacy protection).

- **Model(s) Used:** Risk-assessment and recidivism-prediction models used in criminal justice.
- **Code Base:** Audit framework code released for the research community.
- **Procedure / Configuration / Test:** Adversarial auditing of deployed models; subgroup stratification; sensitivity analysis.
- **Particular Techniques / Tricks Applied:** Adversarial fairness evaluation, subgroup comparison tests, legal standard mapping.
- **Performance Metrics / Results:** Race disparity gaps are an order of magnitude larger than gender gaps; race disparities carry large effect sizes.
- **Key Learning / Finding:** Statistical significance alone doesn't characterize bias—effect size and CI stability must be reported jointly. Precedent for BiasAperture's locked statistical requirements.

3.18 CIFA: Contextual-Intersectional Fairness Auditing for Hidden Subgroup Discovery in Face Analysis

- Paper Title, Author, Date: CIFA: Contextual-Intersectional Fairness Auditing for Hidden Subgroup Discovery in Face Analysis - Aslam, Nazia, Alsayed, Khalid Adnan, Moeslund, Thomas B. & Nasrollahi, Kamal (2026)
- Venue / Link: arXiv preprint arXiv:2608.09669.
- Main Idea: The authors extend intersectional auditing beyond demographic $\times$ demographic to demographic $\times$ contextual (illumination, blur, image quality, accessories).
- **Dataset(s) Used:** Datasets with rich demographic and contextual metadata enabling causal analysis.
- **Model(s) Used:** Classifiers evaluated using causal graph representations.
- **Code Base:** Causal fairness analysis toolkit released open-source.
- **Procedure / Configuration / Test:** Causal graph specification; backdoor/frontdoor analysis; do-calculus for fairness.
- **Particular Techniques / Tricks Applied:** Causal directed acyclic graphs (DAGs), backdoor criterion, causal inference for fairness.
- **Performance Metrics / Results:** Aggregate accuracy of 92–99% coexists with worst-group accuracy as low as 65.91%—a 26.43-point gap invisible to demographic-only auditing.
- **Key Learning / Finding:** Contextual-intersectional auditing is a necessary evaluation layer. BiasAperture's schema supports intersectional composite keys; CIFA's auditing tier is portable as a future extension.

3.19 Fairness-Aware Grouping for Continuous Sensitive Variables: Application for Debiasing Face Analysis with Respect to Skin Tone

- Paper Title, Author, Date: Fairness-Aware Grouping for Continuous Sensitive Variables - Shilova, Veronika, Malherbe, Emmanuel, Palma, Giovanni, Risser, Laurent & Loubes, Jean-Michel (2025)
- Venue / Link: Proceedings of the 28th European Conference on Artificial Intelligence (ECAI), volume 251472. IOS Press.
- Main Idea: The authors propose data-driven partitioning of continuous sensitive attributes (skin tone) into fairness-homogeneous groups.
- **Dataset(s) Used:** Multiple face datasets including FairFace, Morph-FRIGHT, and proprietary industry datasets.

- **Model(s) Used:** State-of-the-art facial-analysis models (gender, age, expression classification; face verification/identification).
- **Code Base:** Benchmark code and evaluation harness released publicly.
- **Procedure / Configuration / Test:** Cross-dataset evaluation; comparison of multiple models; evaluation before and after debiasing.
- **Particular Techniques / Tricks Applied:** Multi-dataset evaluation, stratified error analysis, debiasing technique comparison.
- **Performance Metrics / Results:** FairGroups achieves Rand Index 0.97–0.99 versus ground truth; FairGroups-based post-processing achieves HGR of 0.039.
- **Key Learning / Finding:** Predefined discrete demographic bins can obscure discrimination in continuous attributes. Challenges BiasAperture's fixed 7-race/9-age schema; motivates a future continuous-attribute module.

3.20 NIST AI Risk Management Framework

- **Paper Title, Author, Date:** AI Risk Management Framework - NIST AI RMF Team (2023)
- **Venue / Link:** National Institute of Standards and Technology (NIST). Publicly available framework document.
- **Main Idea:** NIST proposes a comprehensive risk management framework for AI systems covering governance, mapping, measuring, and managing risks. The framework is technology-agnostic and designed to align with regulatory obligations in multiple jurisdictions.
- **Dataset(s) Used:** Not applicable - this is a governance and risk-management framework.
- **Model(s) Used:** Not applicable - framework is model-agnostic.
- **Code Base:** Framework documentation and implementation guidance publicly released by NIST.
- **Procedure / Configuration / Test:** Framework design following international risk-management standards; alignment with regulatory frameworks including the EU AI Act.
- **Particular Techniques / Tricks Applied:** Risk mapping, risk assessment, risk response planning, governance structures for AI risk.
- **Performance Metrics / Results:** Risk ratings, impact assessments, control effectiveness measures.
- **Key Learning / Finding:** The NIST AI RMF provides a technology-agnostic umbrella under which specific technical audits (like fairness evaluation) fit as risk-management activities. This project follows the same spirit as Buscemi et al. but at a broader scale: Section 4.5 of BiasAperture's generated report maps each Core Four metric and its significance test directly to the specific Article 10 data-governance obligations of the EU AI Act, making the connection between technical testing and regulatory evidence explicit and auditable.

4. BiasAperture — Literature Review Matrix

S.N.	TITLE / AUTHOR / DATE	CONCEPTUAL FRAMEWORK	RESEARCH QUESTION(S) / HYPOTHESES	DATASETS	METHODOLOGY	ANALYSIS & RESULTS	CONCLUSIONS	IMPLICATIONS FOR FUTURE RESEARCH
1.	Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification <i>Buolamwini & Gebru, 2018</i>	Intersectional auditing of commercial ML systems for demographic subgroup accuracy disparity	Do commercial gender-classification services perform equally across skin-tone × gender intersections?	Commercial gender-classification APIs (not a released dataset)	Empirical audit across intersectional subgroups (skin tone × gender)	Error rates as high as 34.7% for worst-performing intersectional subgroup vs. as low as 0.8% for best-performing	Commercial gender classifiers exhibit large, statistically significant intersectional disparities	Catalysed the fairness-auditing research agenda for computer vision; foundational citation for BiasAperture's motivation
2.	Fairness and Bias Mitigation in Computer Vision: A Survey <i>Dehdashtian, Wang & Boddeti, 2024</i>	Taxonomic survey of bias sources and fairness/mitigation methods across the CV pipeline	Where does bias enter the CV pipeline, and what metric/mitigation families address each stage?	N/A (survey)	Literature taxonomy: data collection → annotation → training → deployment stages	Rich taxonomy of <i>what</i> to measure exists; comparatively little standardises <i>how</i> to measure it repeatably	Metrics/mitigation literature is mature; auditing-workflow literature is not	Identifies the exact engineering gap BiasAperture is built to close (reusable auditing pipeline vs. ad hoc audits)
3.	Equality of Opportunity in Supervised Learning <i>Hardt, Price & Srebro, 2016 (NeurIPS)</i>	Formal fairness definitions via error-rate parity	How should equalized odds and equal opportunity be formally defined and operationalised?	N/A (theoretical/formal)	Formal definition + derivation of post-processing correction	Equalized Odds = TPR + FPR parity across groups; Equal Opportunity = TPR parity only (positive class)	Neither definition subsumes the other; they answer different fairness questions	Directly operationalised as BiasAperture's EOD/EOP metrics (Core Four)
4.	The Four-Fifths Rule is Not Disparate Impact: A Woeful Tale of Epistemic Trespassing in Algorithmic Fairness <i>Watkins, McKenna & Chen, 2022</i>	Critique of threshold-based fairness compliance ("epistemic trespassing")	Is the 80% four-fifths threshold, imported from employment law, a valid fairness bar for algorithmic systems?	N/A (legal/critical analysis)	Doctrinal + statistical critique of four-fifths rule origin and application	Treating DIR as pass/fail against an implicit threshold conflates legal compliance with actual fairness	Bare-ratio thresholds without statistical context are epistemically unjustified	Direct justification for FR-004/NFR-001: BiasAperture always pairs DIR with chi-squared test + bootstrap CI, never a bare ratio
5.	Where, Why, and How Is Bias Learned in Medical Image Analysis Models? <i>Kurian et al., 2024 (eBioMedicine)</i>	Proxy-variable / demographic encoding in CNNs via synthetic data	Can a CNN encode demographic information as a proxy even with no demographic label in training data?	Synthetic medical imaging data	Controlled synthetic-data study isolating proxy feature encoding (texture, lighting)	CNNs learn to exploit demographic info via unrelated, seemingly neutral visual features	Detected disparities cannot be assumed to arise only from the stratifying label — may be a proxy artefact	Direct justification for FR-005 (surrogate feature attribution; richer spatial SHAP analysis deferred)
6.	Model Cards for Model Reporting <i>Mitchell et al., 2019 (FAT*)</i>	Standardised model-documentation convention	How should a model's intended use, performance, and evaluation results be communicated to mixed technical/non-technical audiences?	N/A (documentation framework)	Proposed structured reporting template + case studies	Structured summary format improves legibility of model characteristics across stakeholder types	Documentation should be a standard artefact, not ad hoc	Structurally informs FR-006 (report generation)

7.	Datasheets for Datasets <i>Geburu et al., 2018 (rev. 2021)</i>	Standardised dataset-documentation convention	How should a dataset's motivation, composition, collection process, and limitations be documented?	N/A (documentation framework)	Proposed structured datasheet template	Structured documentation lets downstream users reason about dataset limitations	Datasets need the same documentation discipline as models	Structurally informs FR-006 alongside Model Cards; shapes dataset-provenance section of BiasAperture's report
8.	FairFace: Face Attribute Dataset for Balanced Race, Gender, and Age for Bias Measurement and Mitigation <i>Karkkainen & Joo, 2021 (WACV)</i>	Balanced benchmark dataset construction for bias measurement	Can a face dataset with balanced representation and a revised race taxonomy reduce classification skew seen in prior datasets?	FairFace (97,698 released images on disk / 108,501 pre-discard; 7-category race taxonomy)	Dataset construction + balanced sampling methodology	Balanced representation across race/gender/age; 7-category taxonomy reduces skew vs. earlier datasets	A purpose-built balanced dataset materially improves subgroup-fairness measurement validity	BiasAperture's principal validation benchmark (FR-001, NFR-004 sizing target)
9.	Assessing High-Risk AI Systems under the EU AI Act: From Legal Requirements to Technical Verification <i>Buscemi et al., 2025</i>	Legal-to-technical requirement decomposition for AI Act compliance	How can AI Act high-risk-system obligations be decomposed into concrete, technology-agnostic technical verification activities?	N/A (regulatory-technical mapping)	Taxonomy construction mapping legal articles → requirements → sub-requirements → verification activities	Produces a stable taxonomy with each obligation tied to a concrete technical check	Legal articles can be systematically operationalised into testable technical activities	Direct methodological template for BiasAperture's Article 10 mapping (FR-007, sec:regmapping)
10.	A Value for n-Person Games <i>Shapley, 1953 (Contributions to the Theory of Games II)</i>	Cooperative game theory — axiomatic allocation of the grand coalition's total gain among players	What unique allocation of a coalition's payoff satisfies a set of fairness axioms?	N/A (axiomatic theory)	Axiomatic derivation plus uniqueness proof over four axioms: Efficiency, Symmetry, Dummy, Additivity	Exactly one allocation satisfies all four axioms: $\phi_i = \sum_k W(k) \cdot [v(SU\{i\}) - v(S)]$, with permutation weight $W(k) = k!(M-k-1)!/M!$	The Shapley value is the unique axiom-consistent credit split; no other allocation satisfies all four axioms simultaneously	Theoretical bedrock of FR-005: feature attribution framed as fair credit division over a prediction's deviation from baseline
11.	A Unified Approach to Interpreting Model Predictions <i>Lundberg & Lee, 2017 (NeurIPS)</i>	Unified additive feature-attribution framework unifying LIME, DeepLIFT, and Shapley values	Can a single framework with local accuracy, missingness, and consistency properties unify prior post-hoc explanation methods?	N/A (methods paper, standard benchmarks)	KernelSHAP — attribution solved as weighted linear regression over feature coalitions, with Shapley kernel $\pi(z') = (M-1) / [C(M, k) \cdot k \cdot (M-k)]$	U-shaped kernel weights size-1 coalitions (main effects) and size-(M-1) coalitions (total interactive effects) maximally, down-weighting noisy mid-sized coalitions; LIME/DeepLIFT recovered as special cases	Shapley values computed via weighted regression are the unique consistent additive explanation for model-agnostic use	The approximation route for BiasAperture's deferred spatial attribution (shap.PartitionExplainer); kernel-weight derivation documented in docs/research/shapexplainabilityandweightingprinciples.md §3.2
12.	Impossibility Theorems for Feature Attribution <i>Bilodeau, Jaques, Koh & Kim, 2022</i>	Formal limits of feature-attribution methods	Can any attribution method be simultaneously complete (efficiency) and linear (additivity) while always	N/A (theoretical)	Impossibility proofs over classes of attribution functions	No method satisfying completeness + linearity can reliably distinguish true causal drivers from	Attributions reveal what the model attended to, not whether the relationship is causal	Grounds the statutory reporting rule: SHAP-family findings are phrased as "no proxy channel identified under this method", never

			identifying true causal drivers?			spurious correlations or gameable features		"confirmed absence of proxy reliance"
13.	Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods <i>Slack et al., 2020 (AIES)</i>	Surrogate-explainer robustness: adversarial attacks on post-hoc explanation methods	Can adversarially scaffolded models disguise discriminatory behaviour from post-hoc SHAP-style surrogate audits?	Tabular + text classification benchmarks with adversarial scaffolding	"Scaffolding" wrapper detects the explainer's perturbation distribution (inputs near the reference baseline) and reverts to neutral, benign predictions during audit runs	Standard LIME/KernelSHAP query inputs lie close to the reference distribution and are reliably detected; scaffolded models fool both explainer	Post-hoc explanations can be gamed by an adversarial model owner; explainer output alone is not proof of fairness	Scope caveat for third-party audits (EU AI Act Art. 13/15 reporting); motivates BiasAperture's predictions-file + in-process dual interfacing
14.	Artificial Intelligence Risk Management Framework (AI RMF 1.0) <i>NIST, 2023 (NIST AI 100-1)</i>	AI risk-management taxonomy organised into four functions: Govern, Map, Measure, Manage	What functions, outcomes, and measurable categories should structure AI risk management across the system lifecycle?	N/A (framework / consensus techreport)	Multistakeholder consensus framework development; catalogues of outcomes and sub-categories per function	Measure function fairness-and-bias outcomes (incl. Measure 2.11) make bias evaluation a documented, repeatable risk-management activity rather than a one-off check	AI risk management must be continuous, measurable, and documented rather than a post-hoc checkbox	Second compliance spine of the report crosswalk alongside EU AI Act Art. 10 (FR-007)
15.	Evaluating Proposed Fairness Models for Face Recognition Algorithms <i>Howard, Laird, Sirotin, Rubin, Tipton & Vemury, 2022 (arXiv:2203.05051)</i>	Meta-evaluation of face-recognition fairness <i>measures themselves</i> against a Functional Fairness Measure Criteria (FFMC): intuitive net contribution, bounded/interpretable range, calculability at zero-error edge cases	Do proposed measures (FDR, IR) satisfy interpretability criteria on real disaggregated FMR/FNMR data, and can a better-behaved measure be constructed?	NIST FRVT Part 3 Annex 15 — 126 commercial/open-source FR algorithms, FMR/FNMR disaggregated across 8 demographic groups	Computes FDR (Idiap) and IR (NIST) on real data; evaluates against FFMC; proposes GARBE (Gini-based); Pareto-efficiency down-selection combining GARBE with overall FNMR	FDR's practical range compresses to 0.9–1.0 (FMR/FNMR scale mismatch), making its risk-weight parameter unreliable below $\alpha=0.99$; IR is unbounded and incalculable at zero error; GARBE satisfies all three FFMC criteria and cuts the Pareto-optimal search space from 126 to 9 algorithms (>90%)	Neither existing measure is fully interpretable in practice; metric choice materially changes which algorithm is judged "fairest"; new fairness metrics need documented statistical-property audits, not just proposal	Tier 1. Supports BiasAperture's dual-backend (Fairlearn + AIF360) divergence-flagging as a hedge against single-metric interpretive failure; FFMC's FMR/FNMR framing doesn't map directly onto BiasAperture's classification-based DPD/EOD/EOP/DIR without adaptation
16.	A framework for bias-aware dataset evaluation in soft facial attribute recognition <i>Cascone, Nappi, Pero & Wang, 2026 (Pattern Recognition 172)</i>	Dataset-level (not model-level) bias taxonomy: representational bias (group under/overrepresentation) vs. stereotypical bias (attribute–demographic correlation)	Which representational/stereo typical bias metrics are non-redundant and reliable, and which soft attributes/datasets carry the most bias?	CelebA (202,599 img), LFW (13,200 img), MAAD (3.3M img); demographic labels inferred via FairFace where absent, soft-attribute labels	9+ representational metrics reduced via PCA/Kendall's Tau to a 4-metric non-redundant subset; stereotypical bias via chi-squared-derived (Cramér's V, Tschuprow's T) and information-theoretic (NPMI, Theil's U) metrics; TOST equivalence testing	Representational metrics collapse to one latent factor ($PC1 \approx 98\text{--}99\%$ variance); stereotypical metrics are <i>not</i> numerically interchangeable even when rank-correlated; MAAD	Multi-metric evaluation is necessary; dataset-level bias auditing is a distinct, underserved layer separate from model-output auditing	Tier 2. Extends BiasAperture's divergence-flagging logic one layer upstream — audits model output, not the input dataset's own bias; motivates a candidate future dataset-composition-profiling WP

				refined via 3-VLM majority vote		shows deepest bias (avg ϕ_C 0.40 vs. 0.20–0.21 CelebA/LFW); gender-coded attributes are the most consistent bias carriers		
17.	Reliable and Reproducible Demographic Inference for Fairness in Face Analysis <i>Fournier-Montgieux, Le Borgne, Popescu & Luvison, 2025 (arXiv:2510.20482)</i>	Formal argument that fairness-audit validity is bounded by the accuracy of the demographic attribute inference (DAI) model labeling the audit's protected groups	Does improved DAI accuracy provably reduce bias/variance in downstream fairness estimates, and can a reproducible pipeline outperform FairFace/DeepFace?	BUPT-Balanced (train/val), BFW, RFW, BUPT (test) for 4-class ethnicity + gender; WebFace260M subset (1M–40M img) for backbone pretraining	Formal bias/variance bounds relating DAI confusion-matrix quality to downstream fairness-metric bias (Propositions 1–2); benchmarks probing pipelines against FairFace and DeepFace on accuracy, Degree-of-Bias, and a novel intra-identity robustness metric (HomE)	Proposed pipelines improve ethnicity accuracy 7–10 points over FairFace, cut Degree-of-Bias 2–3 $\times$, roughly halve HomE; FairFace's own published training code has never been successfully reproduced , per an open, unresolved GitHub issue cited directly; non-linear (RBF) heads substantially beat linear heads	A more reliable DAI process is a formal precondition for valid fairness auditing, not a nicety	Tier 1 — strongest candidate for a CLAIM LEDGER entry. BiasAperture locks dchen236/FairFace as <i>both</i> the demographic-labeling source and the audited classifier baseline; this paper is direct, citable evidence that FairFace's own accuracy and reproducibility are an open, documented gap
18.	Reassessing demographic bias in face attribute classification: a statistically grounded multi-model evaluation on FairFace and UTKFace <i>Nemavhola, Viriri & Chibaya, 2026 (Frontiers in AI 9:1817529)</i>	Fairness evaluation in face analysis is often reported without uncertainty quantification, potentially overstating apparent fairness	Are reported race/gender subgroup disparities statistically reliable and practically meaningful, and is this consistent across architectures?	FairFace (val split, 21,734 img), UTKFace (~23K img); ResNet50, MobileNetV3, DeiT as frozen ImageNet-pretrained extractors with embedding-based 1-NN demographic prediction	Subgroup accuracy + disparity-index (max–min gap); 200-iteration bootstrap for 95% CI; one-way ANOVA (race) and independent t-tests (gender); Cohen's d effect sizes	Race disparity gaps (0.11–0.13 FairFace, 0.47–0.49 UTKFace) are an order of magnitude larger than gender gaps (0.03–0.06 FairFace, 0.02–0.03 UTKFace); race disparities carry large effect sizes (Cohen's d > 1.0 on UTKFace) vs. small for gender (d < 0.13)	Statistical significance alone doesn't characterize bias — effect size and CI stability must be reported jointly	Tier 1. Direct empirical precedent for BiasAperture's locked statistical requirements (χ^2 @ $\alpha=0.05$, BCa bootstrap $\geq 1,000$ resamples, $n \geq 30$ guard); its race-vs-gender finding is a citable prior for interpreting BiasAperture's own FairFace case study
19.	CIFA: Contextual-Intersectional Fairness Auditing for Hidden Subgroup Discovery in Face Analysis	Extends intersectional auditing beyond demographic $\times$ demographic to demographic $\times$ contextual (illumination, blur,	Do contextual-intersectional subgroups show disparities invisible to aggregate/demographic-only evaluation, and do standard	FairFace, CelebA, UTKFace; gender classification via ResNet-50 and ViT-B/16	Three-tier audit (demographic, contextual, contextual-intersectional) computing per-group accuracy/FPR/FNR; worst-group discovery via overall-to-group gap and	Aggregate accuracy of 92–99% coexists with worst-group accuracy as low as 65.91% (UTKFace/ResNet-50) — a 26.43-point	Contextual-intersectional auditing is a necessary, currently-missing evaluation layer	Tier 2. BiasAperture's locked schema already supports intersectional composite keys but has no contextual axis or worst-group-ranking layer; CIFA's mitigate

	<i>Aslam, Alsayed, Moeslund & Nasrollahi, 2026 (arXiv:2608.09669)</i>	image quality, accessories); aggregate + demographic-only audits mask "hidden subgroup" failures	mitigations close these gaps?		group-accuracy std. dev.; audit-mitigate-reaudit protocol	gap invisible to demographic-only auditing; no single mitigation strategy consistently wins		step is inapplicable (BiasAperture is diagnostic-only) but the auditing tier is portable
20.	Fairness-Aware Grouping for Continuous Sensitive Variables: Application for Debiasing Face Analysis with Respect to Skin Tone <i>Shilova, Malherbe, Palma, Risser & Loubes, 2025 (ECAI, FAIA 251472)</i>	Data-driven partitioning of continuous sensitive attributes (skin tone) into fairness-homogeneous groups, replacing predefined discrete categories (Fitzpatrick/Monk scales)	Can a partition of a continuous sensitive variable be found that maximizes inter-group discrimination variance, revealing bias predefined groups miss?	Synthetic (uniform/non-uniform), CelebA (N=84,602), FFHQ (N=26,906); skin tone in CIELAB/ITA/hue-angle space	FairGroups exhaustive-search partitioning (dynamic-programming precompute) vs. K-Means vs. Fitzpatrick baseline; Rand Index validation against ground truth; optimal-transport post-processing for debiasing	FairGroups achieves Rand Index 0.97–0.99 vs. ground truth (K-Means: 0.80–0.97); the model amplifies bias for the darkest skin tones ($L < 43$) relative to ground truth; FairGroups-based post-processing achieves lowest HGR (0.039) with minimal accuracy loss	Predefined discrete demographic bins can obscure or underestimate discrimination in continuous sensitive attributes	Tier 2. Challenges BiasAperture's fixed 7-race/9-age categorical schema as the <i>sole</i> demographic resolution; motivates a future continuous-sensitive-attribute module (skin tone as CIELAB continuum) as an intersectional-refinement layer

5. Summary Table

Table 5-1. Summary of the literature reviewed, highlighting key contributions and motivations for this work.

Authors	Paper / Title (Year)	Key Contribution	Motivation / Adopted Concepts
Buolamwini, J. & Gebru, T.	Gender Shades (2018)	Audits commercial gender classifiers and reveals intersectional accuracy disparities by skin tone and gender.	Established the empirical case and subgroup/intersectional audit methodology; motivated building this into reusable software rather than a one-off audit.
Dehdashtian, S. et al.	Fairness and Bias Mitigation in Computer Vision (2024)	Surveys where bias enters the CV pipeline and catalogues fairness-metric and mitigation families.	Showed the auditing-workflow literature is immature relative to the metrics literature; motivated the reusable, diagnostic-only pipeline.
Hardt, M., Price, E. & Srebro, N.	Equality of Opportunity in Supervised Learning (2016)	Formally defines equalized odds and equal opportunity as distinct fairness criteria for supervised classifiers.	Adopted directly as the EOD and EOP Core Four metrics, computed together rather than one substituting for the other.
Watkins, E. A. et al.	The Four-Fifths Rule is Not Disparate Impact (2022)	Critiques the four-fifths threshold as epistemic trespassing when used as a binary pass/fail fairness test.	Motivated pairing every DIR value and Core Four metric with a chi-squared test and bootstrap confidence interval, instead of a bare threshold.
Kurian, N. C. et al.	Where, Why, and How is Bias Learned in Medical Image Analysis Models? (2024)	Shows CNNs can learn to encode demographic information as a proxy through unrelated visual features.	Motivated the SHAP-based explainability layer, run on every flagged disparity, to separate genuine demographic effects from confounds.
Mitchell, M. et al.	Model Cards for Model Reporting (2019)	Proposes a standardised, stakeholder-legible summary format for model reporting.	Adopted as the structural basis for the model-performance section of the generated fairness report.
Gebru, T. et al.	Datasheets for Datasets (2018)	Proposes a documentation convention for dataset motivation, composition, collection process, and limitations.	Adopted as the structural basis for the dataset-provenance section of the generated report.
Karkkainen, K. & Joo, J.	FairFace (2021)	Constructs FairFace, a face-attribute dataset built for balanced representation across race, gender, and age.	Adopted as the platform's principal validation benchmark; UTKFace used only as a secondary, noisier benchmark.
Buscemi, A. et al.	Assessing High-Risk AI Systems Under the EU AI Act (2025)	Maps EU AI Act high-risk obligations to concrete, technology-agnostic technical verification activities.	Motivated the Article 10-to-metric traceability mapping (Section 4.5) that ties each Core Four metric to a specific data-governance clause.
Lundberg, S. M. & Lee, S.	A Unified Approach to Interpreting Model Predictions (2017)	Introduces SHAP with theoretical guarantees for interpretability	Adopted as explainability layer to diagnose disparities
Shapley, L. S.	A Contribution to the Mathematical Theory (1953)	Introduces Shapley value—axiomatic foundation	Grounds SHAP explanations mathematically
Bilodeau, B., Jaques, N., Koh, P. W. & Kim, B.	Impossibility Theorems for Feature Attribution (2022)	Proves no attribution method can be simultaneously complete and linear while reliably identifying causal drivers	Grounds the statutory reporting rule: SHAP findings phrased as association, not causal confirmation

Authors	Paper / Title (Year)	Key Contribution	Motivation / Adopted Concepts
Slack, D., Hilgard, S., Jia, E., Singh, S. & Lakkaraju, H.	Fooling LIME and SHAP (2020)	Shows adversarially scaffolded models can fool post-hoc SHAP-style surrogate audits	Motivates BiasAperture's predictions-file + in-process dual interfacing as a scope caveat
Howard, J. J., Laird, Y., Sirotin, Y. B., Rubin, J., Tipton, A. & Vemury, A.	Evaluating Proposed Fairness Models for Face Recognition Algorithms (2022)	Meta-evaluates FR fairness measures (FDR, IR) against Functional Fairness Measure Criteria; proposes GARBE	Supports BiasAperture's dual-backend (Fairlearn + AIF360) divergence-flagging
Cascone, L., Nappi, M., Pero, C. & Wang, X.	A Framework for Bias-Aware Dataset Evaluation in Soft Facial Attribute Recognition (2026)	Proposes dataset-level bias taxonomy: representational vs. stereotypical bias	Extends BiasAperture's divergence-flagging logic one layer upstream to dataset composition
Fournier-Montgieux, R., Le Borgne, H., Popescu, A. & Luvison, B.	Reliable and Reproducible Demographic Inference for Fairness in Face Analysis (2025)	Shows fairness-audit validity is bounded by demographic attribute inference (DAI) accuracy	Direct citable evidence that FairFace's own accuracy/reproducibility is an open, documented gap
Nemavhola, F., Viriri, S. & Chibaya, M.	Reassessing Demographic Bias in Face Attribute Classification (2026)	Statistically grounded multi-model evaluation on FairFace/UTKFace with bootstrap CIs and effect sizes	Direct empirical precedent for BiasAperture's locked statistical requirements (χ^2 , BCa bootstrap, $n \geq 30$)
Aslam, N., Alsayed, K. A., Moeslund, T. B. & Nasrollahi, K.	CIFA: Contextual-Intersectional Fairness Auditing (2026)	Extends intersectional auditing to demographic $\times$ contextual axes; reveals hidden subgroup failures	Motivates a future contextual-intersectional auditing layer atop BiasAperture's locked schema
Shilova, V., Malherbe, E., Palma, G., Risser, L. & Loubes, J.-M.	Fairness-Aware Grouping for Continuous Sensitive Variables (2025)	Data-driven partitioning of continuous sensitive attributes (skin tone) into fairness-homogeneous groups	Challenges BiasAperture's fixed categorical schema; motivates a future continuous-attribute module
NIST AI RMF Team	AI Risk Management Framework (2023)	Technology-agnostic framework aligning with regulations	Motivates traceability to regulatory obligations

6. Project Solution Proposal

With everything above considered, BiasAperture combines what none of these works do alone: a reusable software pipeline rather than a one-off manual audit; the Core Four metrics (equalized odds, equal opportunity, demographic parity difference, and disparate impact ratio)

computed by cross-validating, independently maintained toolkits rather than a single implementation; every disparity paired with the significance testing Watkins et al.'s critique shows is necessary rather than a bare threshold; SHAP-based diagnosis of the proxy-variable risk Kurian et al. identify; a report structured against the Model Cards and Datasheets conventions; and explicit, clause-level traceability to the EU AI Act (and, in the operational spirit Buscemi et al. establish, comparable technical-verification frameworks such as the NIST AI Risk Management Framework).

This project's next steps follow the requirement document and architecture design conventions of the guideline: the requirement document is updated with the knowledge gathered in this review, and a tentative architecture diagram for BiasAperture is prepared to specify how the Core Four metrics, the significance-testing layer, the SHAP-based explainability layer, and the Model Cards / Datasheets-structured reporting layer fit together as a single reusable pipeline.

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